

CLASS XII · LECTURE 4

Loops, break and continue

Option I · Two periods · Tables, factorials and number patterns

By the end of this lecture you can

- Choose for, while or do-while and say why.
- Write a multiplication table and a factorial loop.
- Distinguish break from continue.
- Print a number triangle with a nested loop.

Three loops, one idea

A loop repeats a block. `for` is the one to reach for when you know how many times, or when you are counting. `while` tests the condition before the body, so the body may never run. `do-while` runs the body once and then tests, so the body always runs at least once. A `for` loop is an iteration statement, not a selection statement. Selection is `if` and `switch`.

Loop	Test	Use it for
for	Before each pass, after an initialisation	A table from 1 to 10, a pattern of 5 rows
while	Before the body	Reading numbers until the user types a sentinel, when the count is unknown
do-while	After the body	A menu that must be shown at least once

```
#include <stdio.h>

int main(void) {
    int n, i;
    printf("Enter a number: ");
    scanf("%d", &n);
    for (i = 1; i <= 10; i++)
        printf("%d x %d = %d\n", n, i, n * i);
    return 0;
}
```

Program 4.1 Multiplication table. Change only the limit if the question asks for 1 to 10.

```
#include <stdio.h>

int main(void) {
    int n, i;
    long fact = 1;
    printf("Enter a number: ");
    scanf("%d", &n);
    for (i = 1; i <= n; i++)
        fact = fact * i;
    printf("Factorial = %ld\n", fact);
    return 0;
}
```

Program 4.2 Factorial. For input 4 the trace is 1, 2, 6, 24.

Trace for 4, which the hint $4! = 24$ is asking you to understand. Start with `fact = 1`. `i = 1` gives 1. `i = 2` gives 2. `i = 3` gives 6. `i = 4` gives 24. Then `i` becomes 5 and the loop stops. `long` is used because factorials grow quickly. The program as written is for a non-negative `n`. Say that in one line if you have room.

break and continue

DEFINITION break and continue

break leaves the loop at once and continues after it. continue skips the rest of this pass and goes to the next test of the loop. break is an exit. continue is a skip.

```
#include <stdio.h>

int main(void) {
    int i;
    for (i = 1; i <= 5; i++) {
        if (i == 3)
            continue;
        printf("%d\n", i);
    }
    return 0;
}
```

Program 4.3 continue skips 3. The loop still finishes. break would have stopped it.

The output is 1, 2, 4, 5. The number 3 is skipped, not used as a reason to abandon the loop. The multiple-choice line “skip the rest of a loop and carry on from the top” is **continue**. break would print 1 and 2 only.

Nested loops and the two patterns

An inner loop finishes all of its passes for one pass of the outer loop. For a triangle, the outer loop is the row and the inner loop is how many numbers are printed on that row.

```
#include <stdio.h>

int main(void) {
    int row, col;
    for (row = 1; row <= 5; row++) {
        for (col = 1; col <= row; col++)
            printf("%d", col);
        printf("\n");
    }
    return 0;
}
```

Program 4.4 Prints 1 / 12 / 123 / 1234 / 12345.

```
#include <stdio.h>

int main(void) {
    int n;
    printf("Number Square\n");
    for (n = 1; n <= 5; n++)
        printf("%d      %d\n", n, n * n);
    return 0;
}
```

Program 4.5 Number and its square. A single loop is the honest tool. Nesting is not required for this pattern.

The square of 1 is 1, of 2 is 4, of 3 is 9, of 4 is 16, of 5 is 25. If a question insists on nested loops for this second pattern, you can still use Program 4.4’s shape and print row and row * row from the outer loop only. Do not invent an inner loop that does nothing useful.

Checkpoint

Try the questions before you read the model answers. Section B answers on the real paper should stay within about six or seven lines.

Multiple choice

1. A for loop is used for:

- A) Selection
- B) Iteration
- C) Compilation
- D) Linking only

2. Which statement starts the next pass of a loop immediately?

- A) break
- B) continue
- C) return from the operating system
- D) #define

Short answers

S1. Differentiate break and continue.

S2. Write a program that prints the factorial of an inputted number.

Model answers

Cover this section while you attempt the checkpoint. The wording is a full-mark style answer, not the only acceptable wording.

MCQ 1. B. Iteration means repetition. if and switch select.

MCQ 2. B. break leaves the loop. continue skips to the next pass.

S1. break ends the loop at once. The statement after the loop runs next. continue skips the rest of the current pass and starts the next pass of the same loop. In a loop from 1 to 5, continue when i is 3 prints 1 2 4 5. break when i is 3 prints 1 2.

S2. Use the program in this lecture: read n, set fact to 1, multiply fact by i for i from 1 to n, and print fact. For n = 4 the running product is 1, 2, 6, 24. Use long so the result has room to grow.